



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 12

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Program:

```
#include <stdio.h>
```

```
long long mergeAndCount(int a[], int temp[], int left, int mid, int right)
```

```
{
```

```
    int i = left, j = mid + 1, k = left;
```

```
    long long count = 0;
```

```
    while (i <= mid && j <= right)
```

```
    {
```

```
        if (a[i] <= a[j])
```

```
            temp[k++] = a[i++];
```

```
        else
```

```
        {
```

```
            temp[k++] = a[j++];
```

```
            count += (mid - i + 1);
```

```
        }
```

```
    }
```

```
    while (i <= mid)
```

```
temp[k++] = a[i++];
```

```
while (j <= right)
```

```
temp[k++] = a[j++];
```

```
for (i = left; i <= right; i++)
```

```
a[i] = temp[i];
```

```
return count;
```

```
}
```

```
long long mergeSort(int a[], int temp[], int left, int right)
```

```
{
```

```
long long count = 0;
```

```
if (left < right)
```

```
{
```

```
int mid = (left + right) / 2;
```

```
count += mergeSort(a, temp, left, mid);
```

```
count += mergeSort(a, temp, mid + 1, right);
```

```
count += mergeAndCount(a, temp, left, mid, right);
```

```
}
```

```
return count;
```

```
}
```

```
int main()
```

```
{
```

```

int a[] = {2, 4, 1, 3, 5};

int temp[5];

int n = 5;


printf("Name: Dharshana Sivakumar\n");

printf("Count Inversions using Modified Merge Sort\n\n");


printf("Number of Inversions = %lld\n",

      mergeSort(a, temp, 0, n - 1));


return 0;
}

```

Output:

main.c	Run ▶	Output
<pre> 28 return count; 29 } 30 31 int countInversions(int a[], int temp[], int left, int right) 32 { 33 int mid, count = 0; 34 35 if (left < right) 36 { 37 mid = (left + right) / 2; 38 39 count += countInversions(a, temp, left, mid); 40 count += countInversions(a, temp, mid + 1, right); 41 count += merge(a, temp, left, mid, right); 42 } 43 44 return count; 45 } 46 47 int main() 48 { 49 int a[] = {2, 4, 1, 3, 5}; 50 int temp[5]; 51 int n = 5; 52 53 printf("Name: Dharshana Sivakumar\n"); 54 printf("Modified Merge Sort - Count Inversions\n\n"); 55 56 printf("Number of Inversions: %d\n", 57 countInversions(a, temp, 0, n - 1)); 58 59 return 0; 60 } </pre>		<pre> Name: Dharshana Sivakumar Modified Merge Sort - Count Inversions Number of Inversions: 3 </pre>